Every Divisor of an Odd-Index Fibonacci Number

A two-square theorem explaining OEIS A135953

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November 14, 2026

Abstract

OEIS A135953 conjectures that a composite Fibonacci number at a prime index having exactly two distinct prime factors is the product of two sums of two squares. We prove a stronger general statement: every positive divisor of F_n is a sum of two integer squares whenever n is odd. The proof uses the classical identity $L_n^2 - 5F_n^2 = 4(-1)^n$ and the sum-of-two-squares theorem for primes. It also applies to the analogous OEIS conjectures with three, four, or more distinct prime factors. This is an elementary consequence of standard identities; no novelty is claimed.

The OEIS question

OEIS A135953 lists composite Fibonacci numbers F_p with prime index p having exactly two *distinct* prime factors, and conjectures that these numbers can be expressed as products of two sums of two squares. Its program counts distinct factors via the length of the prime factorization; our argument also handles repeated prime powers.

Theorem 1. *For every positive odd integer n, every positive divisor $d \mid F_n$ can be written*

$$\boxed{d = x^2 + y^2 \qquad \text{for some } x, y \in \mathbb{Z}.}$$

More precisely, every odd prime divisor of F_n is congruent to $1 \pmod 4$.

Proof. Define Fibonacci and Lucas numbers by $F_0 = 0$, $F_1 = 1$, $L_0 = 2$, $L_1 = 1$, and the recurrence $u_{n+2} = u_{n+1} + u_n$. The well-known identity

$$L_n^2 - 5F_n^2 = 4(-1)^n \tag{1}$$

can be verified directly: $L_n = 2F_{n+1} - F_n$ and $F_{n-1} = F_{n+1} - F_n$ give

$$L_n^2 - 5F_n^2 = 4(F_{n+1}F_{n-1} - F_n^2) = 4(-1)^n,$$

where the last equality is Cassini's identity (itself immediate from the determinant of the Fibonacci recurrence matrix).

Now suppose n is odd and q is an odd prime divisor of F_n. Reducing (1) modulo q yields

$$L_n^2 \equiv -4 \pmod q, \qquad (L_n\, 2^{-1})^2 \equiv -1 \pmod q.$$

Thus -1 is a quadratic residue modulo q. Euler's criterion, or the fact that an element of order 4 exists in $(\mathbb{Z}/q\mathbb{Z})^\times$, implies $4 \mid q - 1$. Hence $q \equiv 1 \pmod 4$.

Every prime divisor of F_n is therefore either 2 or congruent to $1 \pmod 4$. Fermat's two-square theorem gives $q = a^2 + b^2$ for each prime $q \equiv 1 \pmod 4$, and $2 = 1^2 + 1^2$. The identity

$$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2$$

shows that products, including arbitrary powers, of numbers representable by two squares are again representable. Every divisor of F_n is such a product, proving the result. □

Theorem 2 (The A135953 consequence)**.** *Let p be prime. Suppose F_p has exactly s distinct prime divisors $q_1, \ldots, q_s$, so that $F_p = \prod_{i=1}^{s} q_i^{e_i}$. Then each prime-power factor $q_i^{e_i}$ is a sum of two squares. In particular, when $s = 2$, F_p is the product of two sums of two squares, proving the OEIS A135953 assertion. The same conclusion holds for each analogous fixed-s assertion.*

Proof. The cases $p = 2, 3$ do not occur when $s \geq 2$, since $F_2 = 1$ and $F_3 = 2$. Hence $p \geq 5$ is odd, and Theorem 1 applies separately to each $q_i^{e_i} \mid F_p$. □

Examples, checks, and status

For example $F_{19} = 4181 = 37 \cdot 113$, with $37 = 1^2 + 6^2$ and $113 = 7^2 + 8^2$. Thus F_{19} has the asserted two-factor decomposition. The stronger assertion was checked exactly for every odd $n < 200$ by factoring F_n and testing that its prime divisors lie in $\{2\} \cup \{q : q \equiv 1 \pmod 4\}$.

The prime-factor congruence and the identity (1) are standard; a 2012 online note entitled "2 Prime Factors of F_p" also explicitly records the prime-index conclusion. This document supplies a convenient self-contained argument and does *not* claim a new theorem or priority.

References and disclosure

1. OEIS Foundation, *A135953: Composite prime-index Fibonacci numbers with two distinct prime factors*, https://oeis.org/A135953 (accessed 2026-11-14); compare A135954 and A135955.
2. E. W. Weisstein, "Lucas Number," *MathWorld*, fundamental identity (10), https://mathworld.wolfram.com/LucasNumber.html.
3. "2 Prime Factors of F_p" (online note, 2012), https://sriasat.wordpress.com/wp-content/uploads/2012/12/fibonacci31.pdf; records an earlier prime-index factor congruence.

AI-authorship notice. Primary author (AI): ChatGPT (OpenAI). This AI-generated exposition has been checked computationally on finite samples but has not undergone independent human peer review; the proof, not the samples, establishes its infinite claim.